

MELANIE VIOLA PARTSCH

Dr. phil.

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Modern psychometrics · Machine learning · Structural equation modeling · Model fit evaluation ·
Scale development · Survey methodology

ACADEMIC POSITIONS

since 08/2023	Postdoctoral researcher , Utrecht University (NL), Department of Methodology and Statistics, Faculty of Social and Behavioural Sciences
since 04/2026	<i>In the large-scale research infrastructure project “The Macroscope” funded by the Dutch Research Council (NWO)</i>
02–03/2026	<i>Visiting researcher, University of Basel (CH), Department of Statistics and Data Science</i>
08/2023–01/2026	<i>In the project “Detection and Classification of Misspecifications in Structural Equation Modeling Using Machine Learning” funded by the German Research Foundation (DFG)</i>
01–07/2023	Independent researcher (completion of the dissertation and further research projects)
03–09/2022	Research associate , University of Education Weingarten (DE), Chair of Educational Psychology; project “Contextual and Individual Determinants of Lifelong Learning” (research; contribution to the preparation of a DFG grant proposal)
03/2017–12/2022	Research associate , GESIS – Leibniz Institute for the Social Sciences, Mannheim (DE), Department Survey Design and Methodology
03–12/2022	<i>Questionnaire Design and Evaluation team: validation of standard sociodemographic variables in the project KonsortSWD (National Research Data Infrastructure, NFDI)</i>
03/2017–02/2022	<i>Scale Development and Documentation team: development of scales for social surveys, reviewing and psychometric consulting for the open-access repository ZIS</i>

ACADEMIC EDUCATION

03/2017–12/2022	Doctorate (Dr. phil., summa cum laude) , GESIS Mannheim / RPTU Kaiserslautern-Landau (DE). Dissertation: “Methodological Challenges in the Assessment of Good Character” (defended: November 10, 2023)
09/2013–10/2016	M.Sc. Psychology , University of Mannheim (DE) (final grade: 1.4); master’s thesis at the Chair of Research Methods and Psychological Assessment
10/2010–09/2013	B.Sc. Psychology , Ulm University (DE) (final grade: 1.4); bachelor’s thesis at the Chair of Work and Organizational Psychology

PUBLICATIONS

Articles in Peer-Reviewed Journals

- Partsch, M. V.** & Goretzko, D. (2026). Detecting model misfit in structural equation modeling with machine learning—A proof of concept. *Multivariate Behavioral Research*, 61(1), 1–24. [doi:10.1080/00273171.2025.2552304](#)
- Goretzko, D., & **Partsch, M. V.** (2025). Predicting measurement model misfit with machine learning while accounting for nuisance parameters—An illustration. *Psychological Test Adaptation and Development*, 6, 187–198. [doi:10.1027/2698-1866/a000111](#)
- Partsch, M. V.**, Sterner, P., & Goretzko, D. (2025). A simulation study on the interaction effects of underfactoring and nuisance parameters on model fit indices. *Structural Equation Modeling: A Multidisciplinary Journal*, 32(6), 1000–1015. [doi:10.1080/10705511.2025.2514592](#)
- Goretzko, D., **Partsch, M. V.**, & Sterner, P. (2025). Embrace the heterogeneity in exploratory factor analysis but be transparent about what you do—A commentary on Manapat et al. (2023). *Psychological Methods*. Advance online publication. [doi:10.1037/met0000759](#)
- Schneider, S. L., Palm, L., & **Partsch, M. V.** (2025). CASMIN versus ISCED: Validating standard education variables for German microdata. *KZfSS Kölner Zeitschrift für Soziologie und Sozialpsychologie*, 77(3), 445–477. [doi:10.1007/s11577-025-01019-8](#)

- Partsch, M. V., Olaru, G., & Lechner, C. M.** (2024). Measuring global character dimensions: An ant colony optimization approach toward three core strength scales. *Journal of Personality Assessment*, 106(5), 665–680. doi:10.1080/00223891.2024.2309994
- Stanciu, A., **Partsch, M. V., & Lechner, C. M.** (2024). Basic human values and the adoption of cryptocurrency. *Frontiers in Psychology*, 15. doi:10.3389/fpsyg.2024.1395674
- Partsch, M. V. & Landberg, M.** (2023). Modeling determinants of lifelong learning according to the Theory of Planned Behavior: A proxy-based approach using PIAAC data. *Adult Education Quarterly*, 74(2), 132–151. doi:10.1177/07417136231208951
- Landberg, M., & **Partsch, M. V.** (2023). Perceptions on and attitudes towards lifelong learning in the educational system. *Social Sciences & Humanities Open*, 8(1), Article 100534. doi:10.1016/j.ssaho.2023.100534
- Partsch, M. V., Bluemke, M., & Lechner, C. M.** (2022). Revisiting the hierarchical structure of the 24 VIA character strengths: Three global dimensions may suffice to capture their essence. *European Journal of Personality*, 36(5), 825–845. doi:10.1177/08902070211017760
- Partsch, M. V. & Danner, D.** (2021). Measuring self-control in international large-scale surveys: Development and validation of a four-item scale in English, French, German, Japanese, Polish, and Spanish. *European Journal of Psychological Assessment*, 37(5), 409–418. doi:10.1027/1015-5759/a000618
- Lechner, C. M., **Partsch, M. V., Danner, D., & Rammstedt, B.** (2019). Individual, situational, and cultural correlates of acquiescent responding: Towards a unified conceptual framework. *British Journal of Mathematical and Statistical Psychology*, 72(3), 426–446. doi:10.1111/bmsp.12164
- Nießen, D., **Partsch, M. V., Kemper, C. J., & Rammstedt, B.** (2019). An English-language adaptation of the social desirability–gamma short scale (KSE-G). *Measurement Instruments for the Social Sciences*, 1, Article 2. doi:10.1186/s42409-018-0005-1

Preprints (Under Review)

- Schuhbeck, T. M. B., **Partsch, M. V., Sterner, P., & Goretzko, D.** (2026). *Effect-based assessments of inter-group comparability – the practical impact of measurement non-invariance & differential item functioning on psychological tests and scales*. PsyArXiv. doi:10.31234/osf.io/sjtp3_v1
- Partsch, M. V., Henninger, M., Ximénez, C., & Goretzko, D.** (2026). *Insights from interpretable machine learning into the interplay of model misspecifications, fit indices, and nuisance parameters*. PsyArXiv. doi:10.31234/osf.io/yfx6s_v1
- Partsch, M. V. & Goretzko, D.** (2026). *A machine learning pipeline to classify the type and severity of misspecifications in latent measurement models*. PsyArXiv. doi:10.31234/osf.io/84ftk_v3

Open-Access Survey Scale Documentations

- Partsch, M. V., Schmidt, I., Haehner, P., & Lechner, C. M.** (2023). *VaLiGo—Measuring basic human values and life goals with a nested 30–20–10-item inventory (German version)*. PsyArXiv. doi:10.31234/osf.io/evr7a
- Bluemke, M., **Partsch, M. V., Saucier, G., & Lechner, C. M.** (2021). *Human character in the IPIP: Towards shorter, more content-valid, and cross-culturally comparable IPIP-VIA character strength scales*. PsyArXiv. doi:10.31234/osf.io/k79qf
- Partsch, M. V., Danner, D., & Chernyshenko, O.** (2018). *CCS-based 7-item traditionalism scale (CCS-T-7). Zusammenstellung sozialwissenschaftlicher Items und Skalen (ZIS)*. doi:10.6102/ZIS261

WORKSHOPS & INVITED TALKS

- Goretzko, D. & **Partsch, M. V.** (2026, August 3–7). *Classical and New Approaches to Latent Variable Modeling in Survey Research* [1-Week Workshop]. GESIS Summer School in Survey Methodology 2026, GESIS – Leibniz Institute for the Social Sciences, Cologne, Germany.
- Partsch, M. V.** (2026, June 23). *Herausforderungen bei der Messung psychologischer Konstrukte* [Challenges in the Measurement of Psychological Constructs] [Invited Talk]. Research-Oriented Practical Training for Master Students of Clinical Psychology and Psychotherapy, Goethe University, Frankfurt am Main, Germany.
- Partsch, M. V.** (2026, March 12). *Why We Should Not Blindly Trust Fixed Cutoffs for Model Fit Indices—Insights From Interpretable Machine Learning* [Invited talk]. Society & Choice Research Seminar, University of Basel, Switzerland.
- Partsch, M. V. & Goretzko, D.** (2025, September 21). *Established and Innovative Methods to Evaluate Model Fit in Structural Equation Modeling* [1-Day Workshop]. 18th Biennial Conference of the German Psychological

Society – Personality Psychology and Psychological Diagnostics (DPPD) Section, University of Kassel, Germany.

Goretzko, D., **Partsch, M. V.**, & Sterner, P. (2025, July 22). *New Perspectives on Measurement Invariance Testing* [1-Day Workshop]. XI European Congress of Methodology (EAM 2025), University of La Laguna, Tenerife, Spain.

Partsch, M. V. (2024, November 9). *Detection and Classification of Misspecifications in Structural Equation Modeling Using Machine Learning* [Invited talk]. Workshop of the German research training group Statistical Modeling in Psychology (SMIP), University of Mannheim, Germany.

CONFERENCE PRESENTATIONS & CHAIRED SYMPOSIA

Partsch, M. V., Henninger, M., Ximénez, C., & Goretzko, D. (2026, 21.–24. Juli). Insights from interpretable machine learning into the interplay of model misspecifications, fit indices, and nuisance parameters. In **M. V. Partsch (Chair)**, *Bridging Psychometrics and Modern Data Science: Applications in Machine Learning and Causal Inference* [Symposium]. 91st Annual Meeting of the Psychometric Society (IMPS), Seoul, South Korea.

Partsch, M. V., & Goretzko, D. (2025, September 29–October 1). Introduction and illustration of a new method for detecting and classifying SEM misfit using machine learning. In **M. V. Partsch (Chair)**, *Measurement and Machine Learning* [Symposium]. 17th Conference of the Methods & Evaluation Section of the German Psychological Society (DGPs), Berlin, Germany.

Partsch, M. V., & Goretzko, D. (2025, July 23–25). A machine learning-based workflow for model evaluation and revision in SEM. In **M. V. Partsch (Chair)**, *Measurement and Machine Learning* [Symposium]. XI European Congress of Methodology (EAM 2025), San Cristóbal de La Laguna, Tenerife, Spain.

Partsch, M. V., & Goretzko, D. (2024, September 16–19). Overcoming cutoffs in measurement model fit evaluation—a new machine learning-based approach. In D. Goretzko (Chair), *Investigating Measurement Invariance (MI) and Assessing Model Fit of Measurement Models* [Symposium]. 53rd DGPs Congress / 15th ÖGP Conference, Vienna, Austria.

Partsch, M. V., Sterner, P., & Goretzko, D. (2024, July 16–19). Sensitivity of model fit indices in SEM to underfactoring. In D. Goretzko (Chair), *Advancements in structural equation modeling (SEM)* [Symposium]. 89th Annual Meeting of the Psychometric Society (IMPS), Prague, Czech Republic.

Partsch, M. V., Schmidt, I., Haehner, P., Hanzlová, R., & Lechner, C. M. (2023, September 24–27). VaLiGo—Measuring basic human values and life goals with a nested 30–20–10-item inventory. In B. Rammstedt (Chair), *Assessment of personality, attitudes, and other constructs* [Symposium]. 17th Biennial Conference of the German Psychological Society – Personality Psychology and Psychological Diagnostics (DPPD) Section, Salzburg, Austria.

Partsch, M. V., Bluemke, M., Oлару, G., & Lechner, C. M. (2021, September 12–15). Identifying and measuring three global dimensions of the VIA trait space based on the IPIP-VIA-R. In M. Bluemke & L. Wagner (Chairs), *How can we best describe and measure character in relation to personality?* [Symposium]. 16th Biennial Conference of the German Psychological Society – Personality Psychology and Psychological Diagnostics (DPPD) Section, Ulm, Germany.

Partsch, M. V., Bluemke, M., & Lechner, C. M. (2019, September 16–18). Applying the Bass-ackwards approach to the IPIP-VIA-R inventory to establish the higher-order structure of Peterson and Seligman's 24 character strengths. In M. Bluemke (Chair), *Solutions to methodological challenges in the equivalent assessment of personality traits and values* [Symposium]. 15th Biennial Conference of the German Psychological Society – Personality Psychology and Psychological Diagnostics (DPPD) Section, Dresden, Germany.

Partsch, M. V., Danner, D., & Rammstedt, B. (2018, September 15–20). Der inkrementelle Nutzen einer 30 Items umfassenden RIASEC-Kurzskala in kulturübergreifenden Bevölkerungsumfragen [The incremental value of a 30-item RIASEC short scale in cross-cultural large-scale assessment]. In M. Bluemke (Chair), *Fortschritte der Skalenentwicklung für internationale Vergleichsstudien [Progress in scale development for cross-national studies (large-scale assessment)]* [Symposium]. 51st Congress of the German Psychological Society (DGPs), Frankfurt, Germany.

Partsch, M. V., Lechner, C. M., Bluemke, M., & Saucier, G. (2018, July 2–5). Character strengths: Towards a shorter (and more content-valid) measure of the IPIP-VIA scales. In D. Danner (Chair), *Advances in assessing non-cognitive skills in large-scale surveys* [Symposium]. 11th International Test Commission (ITC) Conference, Montréal, Canada.

- Partsch, M. V., Danner, D., & Rammstedt, B. (2018, July 2–5). Vocational interests: The incremental value of a 30-item short scale suitable for large-scale assessment in six languages. In D. Danner (Chair), *Advances in assessing non-cognitive skills in large-scale surveys* [Symposium]. 11th ITC Conference, Montréal, Canada.
- Partsch, M. V., Bluemke, M., Lechner, C. M., & Saucier, G. (2018, May 3–5). *Toward shorter (and more content-valid) measures of character strengths: Revising the IPIP-VIA scales* [Poster]. 3rd Conference of the German Society for Positive-Psychological Research (DGPPF), Bochum, Germany.
- Partsch, M. V. (2017, September 4–6). *Facettentheoretische Entwicklung und Validierung einer Skala zu Personal Need for Structure [Applying facet theory to develop a Personal Need for Structure scale and its validation]* [Poster]. 14th Biennial Conference of the German Psychological Society – Personality Psychology and Psychological Diagnostics (DPPD) Section, Munich, Germany.

TEACHING & SUPERVISION

Teaching	<i>Advanced Research Methods and Statistics</i> , B.Sc. Psychology, Utrecht University (teaching workgroups) · <i>Multilevel Analysis</i> , research master's programmes, incl. Educational Sciences (computer labs, contribution to written exams, grading) · <i>Introduction to Structural Equation Modeling Using Mplus</i> , Utrecht Summer School (computer labs, consultation for individual research projects, supporting lectures) · <i>Methodology and statistics consultation</i> for bachelor's and master's theses at the Faculty of Social and Behavioural Sciences
Supervision	Four master's theses in the Applied Data Science programme (Utrecht University) · four research internships at GESIS (Scale Development and Documentation team)

ACADEMIC SERVICE

Editorial service	Associate Editor, <i>mda (methods, data, analyses)</i>
Reviewing	Educational and Psychological Measurement · European Journal of Psychological Assessment · Journal of Personality Assessment · Journal of Survey Statistics and Methodology · Measurement Instruments for the Social Sciences · Psychological Test Adaptation and Development
Memberships	DGPs (German Psychological Society) · DPPD Section of the DGPs (Differential Psychology, Personality Psychology and Psychological Assessment)
Talent Development	Invited Mentor for the Society and Choice doctoral program of the Faculty of Psychology at the University of Basel, Switzerland

LANGUAGES & SOFTWARE

Languages	German (native) · English (fluent) · Dutch (intermediate) · Swedish (basic) · French (basic)
Software	R (advanced) · Mplus (intermediate) · SPSS (intermediate) · L ^A T _E X (basic) · HLM (basic)

ACADEMIC REFERENCES

- Prof. Dr. David Goretzko, Professor of Psychological Methods, Goethe University Frankfurt, goretzko@psych.uni-frankfurt.de
- Prof. Dr. Mirka Henninger, Professor of Statistics & Data Science, University of Basel, mirka.henninger@unibas.ch
- Dr. Gerard Saucier, Professor Emeritus of Psychology (Cultural and Moral Personology), University of Oregon, gsaucier@uoregon.edu
- Prof. Dr. Manfred Schmitt, Professor Emeritus of Personality Psychology and Psychological Assessment, RPTU Kaiserslautern-Landau, Manfred.Schmitt@rptu.de